

ATLS 5214

Big Data Architecture

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Electives	AI Engineer (79.0 / 100)	58.7 / 100
SPECIALIZATION On-Campus Elective		

59

CURRENT AVG SCORE
across 6 job roles

96

POTENTIAL SCORE
with recommended additions

+43

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Provides students with a comprehensive survey of technologies used today in the collection, storage, processing, analytics and display of big data. Focuses on cultivating real world skills with students working on semester long projects to execute on a group project.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	79.0	100.0	+21.0	MEDIUM
Applied Scientist	58.8	100.0	+41.2	HIGH
ML Engineer	57.2	94.0	+36.8	HIGH
Machine Learning Engineer	57.2	100.0	+42.8	HIGH
Data Scientist	56.6	100.0	+43.4	HIGH
MLOps Engineer	43.5	84.7	+41.2	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Business Communication 2 of 6 roles

insights

business

Model Deployment & Serving 2 of 6 roles

production

Feature Engineering 1 of 6 roles

preprocessing

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.

3 **MEDIUM** **Consider Model Deployment & Serving content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.

4 **LOW** **Optional: Feature Engineering** — would broaden appeal to one additional role. Candidate terms: preprocessing.

GAP KEYWORDS BY ROLE

AI Engineer

79.0 → 100.0 (+21.0 pts)

following

engineering

learning

build

software

technical

systems

teams

Applied Scientist

58.8 → 100.0 (+41.2 pts)

lab

discovery

scientific

research

literature

interdisciplinary

technical

learning

ML Engineer

57.2 → 94.0 (+36.8 pts)

discovery

learning

machine

systems

production

engineering

starting

build

Machine Learning Engineer

57.2 → 100.0 (+42.8 pts)

engineering

sets

learning

tools

solutions

business

mobile

time

Data Scientist

56.6 → 100.0 (+43.4 pts)

sets

characteristics

evolution

insights

analytical

social

build

engineering

MLOps Engineer

43.5 → 84.7 (+41.2 pts)

production

text

systems

learning

machine

preprocessing

engineering

tools

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.